
Video-in-the-Loop: Span-Grounded Long Video QA with Interleaved Reasoning

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Abstract

We present *Video-in-the-Loop* (ViTL), a two-stage long-video QA framework that preserves a fixed token budget by first *localizing* question-relevant interval(s) with a low-fps skim and then *answering* via span-aware reallocation of visual tokens at higher effective frame rate, emitting an interleaved output with both spans and the final option for direct attribution. We also introduce *VGrounding-QA*, which converts description based event graphs into *span-grounded* multiple-choice QA by pairing each question with *ground-truth* time span(s) and related reasoning. ViTL is trained end-to-end with an interleaved group-relative objective that couples temporal IoU for localization with answer correctness, allowing credit to flow from answers back to spans without increasing compute. Under fixed token budgets, ViTL attains up to 8.6% with 50% less frame input on long-video QA and temporal grounding (e.g., Charades-STA, ActivityNet-Captions) and ablations show that span-aware token reallocation consistently surpasses uniform sampling. Together, *VGrounding-QA* and ViTL provide an interpretable, compute-efficient recipe for scalable long-video QA.

1. Introduction

Multimodal large language models (MLLMs) have advanced rapidly (Hurst et al., 2024; Li et al., 2024a; OpenGVLab Team, 2024), showing strong performance in instruction following, open-vocabulary perception, and multi-step reasoning across images and videos. Recent systems extend temporal context windows, add memory

modules, and leverage stronger backbones (e.g., 3B–70B+ vision language models), making long-form understanding an increasingly realistic goal.

Despite rapid progress, long-video QA remains brittle. Under fixed token and frame budgets, models typically adopt uniform or heuristic sampling (Team, 2025; Lin et al., 2023), spending most capacity on background and overlooking the brief moments that carry the answer. Public training sets (Gao et al., 2017; Fu et al.; Wu et al.) rarely bind each question to *ground-truth* temporal spans, so systems learn to produce answers without reliably learning *where* to look—hindering attribution and limiting evaluation beyond accuracy. Moreover, localization and answering are often optimized in isolation; even when span prediction improves (e.g., higher tIoU), those gains do not consistently translate into better QA because the learning signal does not reward spans that actually improve answers.

Recent efforts extend context windows and adopt adaptive or key-frame selection to reduce background dilution, yet many tokens still land off-target and attribution remains limited. *Temporal Video Grounding* (TVG) methods (Liu et al.; Qu et al., 2024) strengthen localization, but their benefits rarely carry over to QA without span-aware training signals. R1-based post-training (Feng et al.; DeepSeek-AI et al.) improves step-by-step reasoning, but in the absence of span supervision and reward coupling it cannot teach *where* to search. These observations motivates us to explore a pipeline that reallocates visual tokens to evidence at fixed cost (zoom-in the video), a dataset that jointly supervises spans and answers, and a learning objective that couples localization quality with answer utility.

We present *Video-in-the-Loop* (ViTL), a two-stage procedure that allocates computation where it matters while preserving a fixed token budget. The model first performs a low-frame-rate *skim* over the entire video to localize one or more evidence intervals, then *zooms* into the predicted spans and reasons at a higher effective frame rate to produce the multiple-choice answer. Both the localized spans and the final option are emitted in a single, interleaved response, yielding direct attribution and a uniform format for evaluation. By design, ViTL turns long-video QA into a skim→zoom workflow that concentrates tokens on evidence

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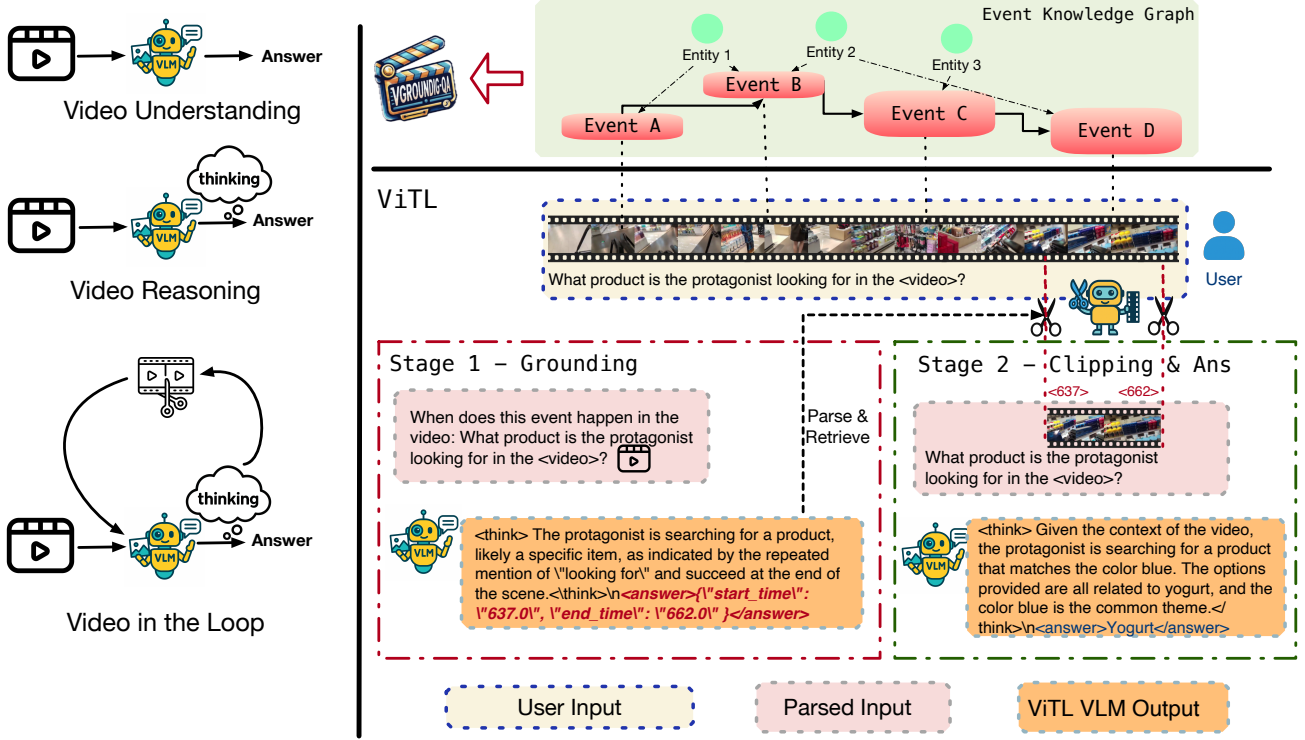


Figure 1. Overview of ViTL (Video-in-the-Loop) and VGrounding-QA. ViTL (right-down): Given a long video V and a question q , **Stage 1 (Ground)** takes a *grounding query* distilled from q (“locate the moments needed to answer q ”) and predicts one or multiple relevant temporal spans $S = \{[t_s^{(i)}, t_e^{(i)}]\}$. Supervision comes from event-graph gold spans. **Stage 2 (Answer)** re-encodes only frames within S at higher fidelity (e.g., higher frame rate/resolution) and answers the original MCQA. Training follows an RL-style loop that jointly optimizes grounding (IoU-based) and QA (cross-entropy or reward) objectives, encouraging spans that improve answering. VGrounding-QA (right-top): The spanning aware training set is achieved from Event Knowledge Graph.

rather than background (see Fig. 1).

We train ViTL end-to-end on the interleaved output using a group-relative policy objective (GRPO) that couples a temporal-IoU signal for span quality with an answer-correctness signal for utility. This composite reward assigns credit from the answering step back to the localization step, so spans are optimized not merely to match annotations but to improve downstream answers under the same compute. A brief supervised warm-up stabilizes decoding, after which interleaved GRPO refines both stages jointly and encourages the model to produce well-formed spans and faithful answers in one pass.

However, in long-video understanding tasks, accurately identifying the time segments relevant to a given question is highly challenging, since many questions are complex, may span multiple temporal segments, and often involve intricate temporal and object relationships. To address this, we innovatively apply *event knowledge graph* techniques: powerful MLLMs such as GPT-4o are first employed to perform fine-grained structured analysis of the entire video, extracting objects, events, and their relationships to construct an event knowledge graph. From this graph, we then select valuable nodes and edges to formulate QA questions.

In this way, we can design multi-hop reasoning questions that span multiple temporal segments, which enables us to train models to simultaneously improve both temporal localization of question-relevant segments and the ability to handle complex long-video understanding problems. (see Fig. 2)

Our contributions can be summarized as follows:

- (1) We introduce a **Video-in-the-Loop** procedure that re-locates tokens to predicted evidence under a fixed budget, producing interpretable span grounding and answers.
- (2) We develop a span-grounded training set construction approach and a training set called VGrounding-QA that ties each QA item to *ground-truth* temporal spans, supplying the missing supervision for span-aware QA.
- (3) We perform end-to-end training that links temporal IoU and answer reward, providing direct credit assignment from answers back to spans.
- (4) We evaluate on **long-video QA** and **temporal grounding**, and conduct ablations to demonstrate the effectiveness of the proposed paradigm.

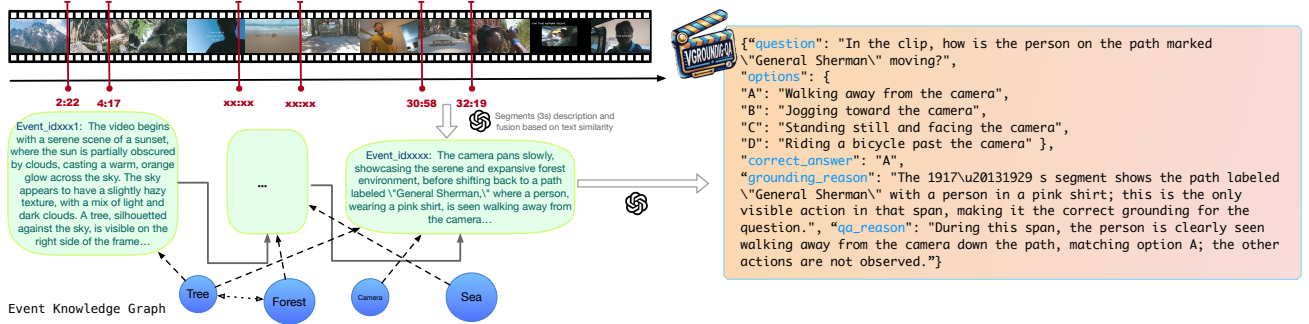


Figure 2. **Training-set construction from event graphs via semantic chunking.** A long video is first buffered into short uniform chunks (e.g., 3s) and produces per-chunk descriptions. Neighboring chunks with high textual similarity are merged into *semantic segments*; their absolute start/end times become the **ground-truth** span(s) (example: 30:58 → 32:19). Each segment is summarized into an event description and converted into a *span-grounded* MCQA instance whose question is answerable using only this span; distractors are mined from other events in the same video.

2. Related Work

2.1. MLLMs for Long-Video Understanding

Recent multimodal LLMs extend language reasoning to video by enlarging temporal context, compressing visual tokens, and adding spatio-temporal adapters. LLaMA-VID reduces per-frame tokens to support hour-scale inputs (Li et al., 2024b); VideoLLaMA2 introduces specialized temporal connectors and an audio branch for richer dynamics (Cheng et al., 2024); and LongVA pushes sequence length into the hundred-thousand-token regime for untrimmed videos (Zhang et al., 2024b). Concurrently, video-tuned backbones (e.g., Qwen2.5-VL; Video-LLaVA) demonstrate strong zero-/few-shot results on captioning and QA (Team, 2025; Lin et al., 2023). Reasoning-centric post-training has also been explored: rule-based RL improves step-by-step solutions in text (Guo et al., 2025) and has been adapted to video settings (Feng et al., 2025). On the localization front, *temporal video grounding* integrates language with segment prediction and is commonly evaluated on Charades-STA and ActivityNet-Captions (Gao et al., 2017; Krishna et al., 2017); recent LLM-based approaches (Ren et al., 2023; Huang et al., 2024; Qian et al., 2024; Zhao et al., 2024) report stronger tIoU/Recall. While effective at broad coverage and zero-shot generalization, many systems still operate over globally sampled frames and rely on training protocols that optimize grounding and answering separately, which can leave attribution and span-to-answer transfer underexploited.

2.2. Multi-Stage and Adaptive Video Processing

A complementary line reduces redundancy by selecting key frames or segments before reasoning. Early systems rely on uniform sampling or CLIP-style retrieval, while recent methods use adaptive temporal focusing to revisit promising clips. VideoZoomer (Ding et al., 2025) and LOVE-R1 (Fu et al., 2025) are especially relevant: both follow a coarse-

to-fine zoom-in paradigm where a model first inspects a low-fidelity view and then requests higher-fidelity evidence from selected temporal regions.

While effective, CLIP-based selection inherits a query-embedding mismatch: interrogative questions are not the distribution CLIP was trained on, so retrieved frames maximize caption-style alignment rather than evidential sufficiency for the posed question. Moreover, top- k retrieval over individual frames tends to break temporal continuity—dropping onsets/offsets and interstitial motion—thereby introducing “negative” frames and missing multi-span evidence crucial for actions and causality. Because the selector is optimized with proxy salience objectives and sits upstream of answering, utility cannot flow back to refine selection, which makes performance sensitive to thresholds and limits temporal attribution.

3. VGrounding-QA: Training Set Creation from Event Graphs

Overview. We construct a span-grounded training set that couples *verifiable temporal supervision* with *answer supervision* as exemplified in Figure 2. Our pipeline proceeds in three stages: (i) selecting source annotations with broad coverage, (ii) converting each annotated event into a single MCQA instance that is answerable from its **ground-truth** temporal span(s), and (iii) performing layered quality review to ensure temporal locality and item difficulty. Table 3 shows a brief comparison with existing long video understanding datasets.

Event Knowledge Graph construction. We adopt event graphs built by a *semantic chunking* pipeline from prior work (Yan et al., 2025). Instead of uniform, fixed-window segmentation alone, a long video is first buffered into short, uniform chunks (e.g., ~ 3 s). A lightweight VLM (e.g., a 7B variant) then produces a brief description for each chunk.

Table 1. Overall statistics of *VGrounding-QA*.

Statistic	Value
Video sources	Video-MME and CG-Bench
Task format	Span-grounded MCQA
Total QA instances	5,240
Unique videos	1,612
Multi-span prevalence	28.4%
Mean span length	24.8s
Human audit pass rate	92.0%

Neighboring chunks are compared via a text-similarity signal (BERTScore over the per-chunk descriptions); adjacent chunks with high similarity are *merged* into a single, temporally contiguous *semantic chunk*, while boundaries are enforced where similarity drops below a threshold. This yields event-level segments that better match the variable temporal granularity of real videos and remain efficient to construct under tight compute. Each semantic chunk is further summarized to obtain a concise event description, and its absolute start/end times define the event’s **ground-truth** span(s). Edges between events are derived from interval relations on these spans (e.g., *Before*, *Overlaps*). Entities are retrieved and linked from each events with semantic de-duplication through embedding based clustering. Entities within the same cluster are then linked.

Conversion to span-grounded QA. For every event node, we first inherit and normalize its span set: timestamps are converted to seconds, spans are sorted and minimally merged when slightly overlapping, invalid ranges are removed, and disjoint occurrences are retained as multi-span. We then distill a short, entity/attribute-aware *grounding query* from the node description using GPT-o3 API (Hurst et al., 2024); deictic phrasing (“this clip/moment”) is avoided so the text remains globally localizable on the timeline. Finally, we synthesize exactly one multiple-choice question (four options) conditioned on the event and its span(s). The question must be answerable using only the annotated evidence (the union when multi-span), and the three distractor options are drawn from other events in the *same video* to provide realistic, in-domain confounds. Brief rationales for span sufficiency (Stage 1) and option correctness (Stage 2) may be retained for supervised learning input.

Data provenance and contamination control. *VGrounding-QA* is a derived training resource built from Video-MME and CG-Bench videos using event-graph parsing and semantic chunking. We split data by video to avoid leakage from shared footage. Since Video-MME contributes source videos to *VGrounding-QA*, we do not report Video-MME as an evaluation benchmark. All reported LongVideoBench, LVBench, MLVU, Charades-STA, ActivityNet-Captions, ActivityNet-CD, QVHighlights,

Table 2. Minimal schema for each training instance. GT = ground truth.

Field	Role
video_id,	Link to the source video and event node
event_id	Supervision for Stage 1 grounding (IoU/Recall); single or multi-span
time_spans (GT)	Human-readable summary of the event node
event_description	Span-seeking reformulation for Stage 1 localization
grounding_query	Prompt for Stage 2 reasoning on the clipped segment(s)
question	Supervision for Stage 2 answer selection
options (A–D),	Optional signals (span sufficiency; option justification)
correct_answer	
stage1_reason,	
stage2_reason	

MVBench, EgoSchema, and EgoTempo results use their official public evaluation splits rather than held-out samples from *VGrounding-QA*.

Quality review. We apply a compact but strict review stack designed to enforce *temporal faithfulness* and reduce annotation leakage. (1) **Schema validity** ensures exactly one correct option and fully parsable fields. (2) **Temporal-locality enforcement** rejects items whose resolution requires frames outside the annotated span(s), or depends on video-level summaries rather than localized evidence; for multi-span events, the question must genuinely require the union of annotated spans when the narrative is distributed across segments. (3) **Language screening** removes deictics and vague stems, keeping questions specific, grounded, and video-dependent. (4) **Text-only leakage filtering** identifies items that can be reliably answered without watching the video; such items are revised or discarded. (5) **De-duplication and bias control** removes near-duplicates within a video and balances option labels to mitigate positional bias. Finally, we conduct a human audit on a stratified sample of instances to directly assess timestamp faithfulness—i.e., whether the annotated span(s) fully cover the visual evidence needed to answer the question. The audit yields a **92% pass rate**, providing direct validation that the temporal supervision is faithful and usable for span-grounded training.

Record schema. Each instance minimally contains the fields in Table 2. Stage 1 uses the *grounding query* and *ground-truth spans* for temporal localization; Stage 2 uses the clipped segment(s) together with *question*, *options*, and *correct answer* for answer supervision.

Splits and reporting. Data are split *by video* to prevent leakage from shared footage. We preserve domain, duration, multi-span prevalence, and option-label balance across

Table 3. Comparison of long-video resources. “MCQA” = multiple-choice QA; “Multi-span” = multiple disjoint spans; “Reason” = per-sample reasoning fields.

Dataset	GT spans	MCQA	Multi-span	Event-graph	Reason
Charades-STA	✓				
QVHighlights	✓		✓		
Ego4D-NLQ	✓		✓		
ActivityNet-QA		✓			
VideoMME		✓			
VGrounding-QA	✓	✓	✓	✓	✓

splits. Aggregate dataset statistics are summarized in Table 1; additional per-split counts, duration distributions, span-length percentiles, and per-video instance counts are reported in the Appendix. The resulting training set pairs **verifiable spans** for temporal grounding with **multiple-choice supervision** for answering. Same-video distractors make localization consequential under a fixed token budget, aligning the data directly with the two-stage protocol in Sec. 4.

4. ViTL: Interleaved Two-Stage GRPO with Grounded Spans

4.1. Two-Stage Video-in-the-Loop with Frame-Level Timestamp Injection

Formulation. Given a long video V of duration $|V|$ and a question Q , the model predicts (i) a set of temporal segments $T = \{[t_s^{(m)}, t_e^{(m)}]\}_{m=1}^M$ that contain the necessary evidence, and (ii) a multiple-choice answer $A \in \{A, B, C, D\}$. Training/evaluation uses *ground-truth* spans $\mathcal{T}^*$ inherited from the event graph (Sec. 3); we allow $M \geq 1$ (multi-span).

Stage 1: Global temporal localization. We sample a *global* sequence of n_g frames uniformly over $[0, |V|]$ to obtain $V_g = \{(x_f, t_f)\}_{f=1}^{n_g}$, where t_f is the absolute time (in seconds) and x_f is the image token. Conditioned on Q and a short grounding query, the model outputs a structured set of segments

$$T = \{[t_s^{(m)}, t_e^{(m)}]\}_{m=1}^M, \quad 0 \leq t_s^{(m)} < t_e^{(m)} \leq |V|,$$

together with a brief rationale. We permit M to vary up to $M_{\max}$; disjoint segments are encouraged when the evidence is non-contiguous.

Stage 2: Span-conditioned answering. Let $U(T)$ denote the (ordered) union of predicted segments. We clip V to $U(T)$ and sample a *local* sequence of n_ℓ frames at higher effective fps, yielding $V_\ell = \{(x_f, t_f)\}_{f=1}^{n_\ell}$ with the *same absolute timestamps* t_f (no re-zeroing). Conditioned on (Q, T, V_ℓ) the model outputs the final option A and a short justification. This reallocates visual tokens from background to evidence while keeping the total budget $n_g + n_\ell$ fixed.

Frame-level timestamp injection (textual). To stabilize temporal reference and enable auditable spans, each frame is serialized as an image token followed by a human-readable absolute time:

$$\langle \text{<image> @ } t_1s, \text{<image> @ } t_2s, \dots, \text{<image> @ } t_{FS}s \rangle$$

We require Stage 1 to emit spans as $\text{}[t_s, t_e]\text{}$ and Stage 2 to answer *using only* frames whose timestamps lie within $U(T)$. Ablations in Sec. 5.4 show that this textual timestamping improves tIoU/Recall and reduces off-by-segment errors under the same token budget.

Budgets and sampling policy. Unless otherwise specified, we use n_g uniformly spaced frames for Stage 1 over $[0, |V|]$ and n_ℓ frames for Stage 2 drawn from $U(T)$ (with per-span caps to prevent degenerate allocation). Segments in T are sorted and minimally merged prior to clipping; multi-span inputs are concatenated in temporal order.

Validity constraints. Outputs are lightly constrained at the prompt level: Stage 1 must place seconds inside $\text{}\dots\text{}$ with numeric values (two decimals), and Stage 2 must emit exactly one option inside $\text{<answer>}\dots\text{</answer>}$. During training, malformed or out-of-range spans are rejected by format checks; at evaluation they are treated as invalid.

4.2. Learning with Interleaved Group-Relative Policy Optimization

GRPO on interleaved sequences. We optimize π_θ with Group-Relative Policy Optimization (GRPO) on interleaved outputs. For each (Q, V) we sample k responses $\{S_i\}_{i=1}^k$, each $S_i = [T_i; A_i]$, and compute group-relative advantages $A_i = R_i - \frac{1}{k} \sum_{j=1}^k R_j$. The objective maximizes the likelihood of higher-reward sequences:

$$\mathcal{L}_{\text{GRPO}} = -\mathbb{E}_{(Q,V)} \sum_{i=1}^k A_i \log \pi_\theta(S_i | V_{\text{unified}}, Q). \quad (1)$$

Coupling both stages within S provides direct credit assignment from answer utility back to localization.

Composite reward with span utility. Each response $S = [T; A]$ is scored by

$$R(S) = (1 - \gamma) R_{\text{loc}}(T) + \gamma R_{\text{ans}}(A), \quad \gamma \in [0, 1]. \quad (2)$$

Localization uses a multi-span temporal IoU against ground truth plus a small format component:

$$R_{\text{loc}}(T) = (1 - \alpha) \text{tIoU}(T, \mathcal{T}^*) + \alpha \text{Fmt}_{\text{time}}(T),$$

where $tIoU$ is the ratio of total intersection to total union length over time; Fmt_{time} rewards in-range, ordered, well-formed timestamps. The answer term rewards exact match and stable formatting:

$$R_{ans}(A) = (1 - \beta)\mathbb{I}[A = A^*] + \beta Fmt_{ans}(A).$$

Small α, β stabilize learning without outweighing task reward. Invalid or unparsable spans receive near-zero Fmt_{time} .

Initialization, schedule, and controls. We initialize from a base VLM (e.g., Qwen2.5-VL-3B/7B). A short supervised warm-up on MCQA *clipped to ground-truth spans* stabilizes decoding. We then run interleaved GRPO with group size k (e.g., $k=3$), per-batch reward normalization, and KL control to the base policy. A light curriculum gradually increases γ from localization-heavy to answer-balanced over early epochs. Full hyperparameters appear in Sec. 5.1.

5. Experiments

5.1. Setup

Long-video QA benchmarks. We evaluate on three public QA suites with multi-choice format and long, open-domain videos: **LongVideoBench** (Val split), **LVBench** (Val), and **MLVU** (Dev), where we report accuracy (%). For MLVU we additionally report the macro average **M-Avg** across tasks.

Temporal Video Grounding (TVG) benchmarks. We use **Charades-STA** (indoor activities; trimmed queries on untrimmed videos) and **ActivityNet-Captions** (open-domain activities). Following standard practice, we report **Recall@IoU={0.3, 0.5, 0.7}** and **mIoU** (%).

Role of our event-graph dataset. Our event-graph grounded dataset (Sec. 3) provides *training* and *diagnostic* supervision: each question is paired with *gold* time span(s) derived from an event graph, enabling learnable temporal grounding and auditable evaluation. Unless otherwise specified, this dataset is not used as a held-out test set; all reported results use the official public splits of the benchmarks above.

Evaluation protocol and fairness. To ensure comparability, all systems run under matched compute budgets. We report (when applicable) *frames* per question and keep the total token/FLOPs budget fixed across baselines. Unless stated, we do not use external subtitles/ASR. Inference for ViTL is two-stage: a low-fps global sweep for *localization* (Stage 1), followed by span-aware high-fidelity *answering* (Stage 2). Results are reported on $4 \times A100$ 80G.

Backbones. We instantiate ViTL with **Qwen2.5-VL 3B** and **Qwen2.5-VL 7B**. Unless otherwise noted, Stage 1 and

Stage 2 share the same backbone. In a compute-efficient setting, we also study a light *selector* (3B) with a stronger *answerer* (7B) in Sec. A.

Training protocol. We adopt a simple three-step schedule that closes the loop between localization and answering while keeping budgets explicit. **(T1) Supervised warm-up.** *Grounding SFT*: train Stage 1 with IoU-based span supervision on our event-graph dataset (gold span(s) per question); we minimize a boundary/IoU loss. *QA SFT*: train Stage 2 on the paired MCQA (cross-entropy over options), using only frames within the *gold* spans. For TVG benchmarks, we do not fine-tune on their train splits unless explicitly marked as FT. **(T2) Two-stage coupling (teacher forcing).** We connect the stages by feeding Stage 1 predictions into Stage 2. To stabilize learning, we sample a fixed ratio of teacher-forced spans (gold) and model spans (predicted) during training, and restrict Stage 2’s visual budget to the union of selected span(s). **(T3) Interleaved GRPO post-training.** We optimize the interleaved span-and-answer sequence with GRPO. For each video-question pair, we sample $k=3$ candidate responses and score each response with a composite reward: *TVG reward* = temporal IoU with the gold span(s), and *QA reward* = 1 for the correct option and 0 otherwise, plus mild format penalties for parsable spans and answers. Group-relative advantages are normalized within each candidate group, and a KL penalty keeps the policy close to the SFT reference. We use a curriculum that starts localization-heavy and gradually moves to an answer-balanced reward, so Stage 1 learns spans that improve Stage 2 accuracy rather than only matching timestamp annotations.

Inputs and budgets. Stage 1 consumes a fixed number of frames (e.g., 64 frames) and takes the global view of the full video and a *grounding query* distilled from the question (“locate the moments needed to answer q ”). Stage 2 re-encodes only the predicted span(s) at higher fidelity (e.g., 4–8 fps and/or higher resolution), up to K spans (default $K=5$), keeping the *total* token/frame budget matched to baselines. Unless noted, Stage 2 answers the original MCQA (no extra hints) on the clipped segment(s).

5.2. Long-Video QA Performance

We evaluate the full ViTL pipeline on **LongVideoBench** (Val), **LVBench** (Val), and **MLVU** (Dev). Unless noted, comparisons follow the no-subtitles, matched-preprocessing setting (global low-fps sweep; fixed resolution) for fairness. Our improvements stem from two ingredients: (i) an *event-graph* dataset that pairs reasoning-centric questions with **ground-truth** time spans (Sec. 3), enabling learnable and auditable temporal grounding; and (ii) a *two-stage, R1-style* training protocol (Sec. 4) that couples IoU-based grounding signals with QA objectives, encouraging spans that *actually* improve answering under fixed token/frame budgets.

Table 4. **Long-video QA benchmarks.** Accuracy (%) on **LongVideoBench** (Val), **LVBench** (Val), and **MLVU** (M-Avg). “Frames” is the per-question frame budget when available. [†] official numbers; [‡] our re-test under the matched preprocessing (global low-fps sweep; uniform sampling; 448 resolution). “—” indicates not reported under the same setting.

Models	Size	Frames	LongVideoBench	LVBench	MLVU M-Avg
<i>Closed Video MLLMs</i>					
GLM-4V-Plus [†]	—	256	70.8	58.7	—
GPT-4o [†]	—	384	66.7	27.0	64.6
Gemini-1.5-Pro [†]	—	0.5 fps	64.0	33.1	—
<i>Small Video MLLMs</i>					
VITA-1.5	7B	16	56.1	—	—
LLaVA-Video	7B	64	58.2	—	—
LongVA	7B	128	52.1	39.4	52.0
NVILA	8B	256	57.7	—	—
ByteVideoLLM	14B	256	—	—	—
VideoLLaMA3	17B	180	59.8	45.3	—
InternVL3	8B	128	62.8	45.6	—
Qwen2.5-VL [†]	7B	256	56.0 (224 res)	45.3	54.5
Qwen2.5-VL [‡]	7B	256	61.8 (448 res)	43.7	—
ViTL (Qwen2.5-VL)	7B	128	63.3	47.4	62.3

Table 5. Zero-shot temporal grounding on Charades-STA (Gao et al., 2017). **Bold** marks our scores.

Method	Size	R@0.3 (%)	R@0.5 (%)	R@0.7 (%)	mIoU (%)
VTimeLLM (Huang et al., 2024)	13B	55.3	34.3	14.7	34.6
TimeChat (Ren et al., 2023)	7B	51.5	32.2	13.4	—
Momentor (Qian et al., 2024)	7B	42.6	26.6	11.6	28.5
HawkEye (Zhao et al., 2024)	7B	50.6	31.4	14.5	33.7
ChatVTG (Qu et al., 2024)	7B	52.7	33.0	15.9	34.9
VideoChat-TPO (Yan et al., 2024)	7B	58.3	40.2	18.4	38.1
E.T. Chat (Liu et al., 2024b)	4B	65.7	45.9	20.0	42.3
ViTL (Ours 7B)	7B	77.7	63.5	36.3	54.0

As shown in Table 4, the largest gains appear on **LVBench**, where relevant moments are sparse and uniform sampling wastes budget. By learning spans from event-graph supervision and coupling them to answering with a two-stage R1-style objective, *ViTL* reallocates visual tokens toward evidence—achieving higher accuracy at comparable (or lower) frame budgets. An oracle-span analysis in Sec. 5.4 further quantifies the remaining headroom from localization fidelity, supporting both the dataset design and the training protocol.

5.3. Temporal Grounding Performance

We then assess *ViTL*’s ability to *localize* question-relevant moments. All results are zero-shot unless noted. The gains are primarily driven by two factors: (i) our *event-graph* dataset that pairs reasoning-centric questions with *groundtruth* time spans (Sec. 3), providing auditable supervision for temporal grounding; and (ii) our *two-stage, R1-style* training that jointly optimizes an IoU-based grounding signal and a QA objective, encouraging spans that improve answering (Sec. 4.2).

Charades-STA (Gao et al., 2017). This benchmark focuses on indoor activities with natural language queries. As shown

in Table 5, **ViTL (Ours 7B)** surpasses specialized VTG systems across recall thresholds and mIoU. **ActivityNet-Captions (Krishna et al., 2017).** Results on open-domain, untrimmed videos are shown in Table 6. **ViTL (Ours 7B)** maintains strong localization across R@IoU levels.

5.4. Ablation Studies on Long-Video QA

We quantify which ingredients drive gains under *matched token/frame budgets* and a fixed backbone (Qwen2.5-VL 3B/7B). We report accuracy (%) on **LongVideoBench** and **LVBench**. All single-stage variants consume $n_g + n_\ell$ frames uniformly over the full timeline; the two-stage system uses n_g (global skim) + n_ℓ (zoomed evidence).

Settings. A) **SFT** — single-stage supervised fine-tuning on our MCQA (no timestamps). B) **SFT + TI** — single-stage with *frame-level textual timestamp injection*. C) **SFT + Stage-2-only + TI (full video)** — no Stage 1; apply the Stage-2 answering prompt with timestamps to the *entire video* (no cropping); total frames $n_g + n_\ell$ sampled uniformly. D) **ViTL(full)** — two-stage with interleaved GRPO (coupled QA+TVG rewards) and timestamp injection in both stages.

Table 6. Zero-shot temporal grounding on ActivityNet-Captions (Krishna et al., 2017). FT indicates fine-tuning on the downstream training split. **Bold** marks our scores.

Method	Size	FT	R@0.3 (%)	R@0.5 (%)	R@0.7 (%)	mIoU (%)
2D-TAN (Zhang et al., 2020)	—	✓	60.4	43.4	25.0	42.5
MMN (Zhang et al., 2021)	—	✓	64.5	48.2	29.4	46.6
VDI (Luo et al., 2023a)	—	✓	—	48.1	28.8	—
VideoChat (Li et al., 2023)	7B	×	8.8	3.7	1.5	7.2
Video-LLaMA (Zhang et al., 2023)	7B	×	6.9	2.1	0.8	6.5
Video-ChatGPT (Maaz et al., 2023)	7B	×	26.4	13.6	6.1	18.9
Valley (Luo et al., 2023b)	7B	×	30.6	13.7	8.1	21.9
ChatVTG (Qu et al., 2024)	7B	×	40.7	22.5	9.4	27.2
Momentor (Qian et al., 2024)	7B	×	42.9	23.0	12.4	29.3
E.T. Chat (Liu et al., 2024b)	4B	×	24.1	12.8	6.1	18.9
ViTL (Ours 7B)	7B	×	55.1	46.3	30.0	24.1

Table 7. Ablations on long-video QA (fixed token budget). Accuracy (%). Higher is better.

Config	LongVideoBench ↑	LVBench ↑
A) SFT (single, no TI)	61.3	40.2
B) SFT + Timestamp Injection (single)	62.1	44.3
C) SFT + Stage-2-only + TI (full video; no crop)	62.5	45.4
D) ViTL(full) : two-stage + GRPO (QA+TVG) + TI	63.5	47.9

Findings. As shown in Table 7: *Timestamping improves global reasoning*. A→B isolates frame-level time tokens in a single-stage setup and yields consistent gains on both benchmarks, indicating reduced temporal ambiguity when evidence is dispersed. *Answer-centric training helps without localization*. B→C shows that the Stage-2 answering prompt with timestamps—applied to the full video at the same budget—further boosts accuracy, suggesting better temporal utilization despite no cropping. *Full ViTL is best at fixed compute*. C→D adds learned localization and coupled QA+TVG rewards via interleaved GRPO. Accuracy improves on both datasets, confirming that allocating tokens to predicted evidence (while keeping totals fixed) yields better answers than any single-stage alternative.

5.5. Effectiveness of Learned Grounding

To validate the necessity of a learned grounding module, we compare *ViTL* against heuristic baselines and establish performance bounds. **Parametric vs. Non-Parametric Frame Selection.** We compare *ViTL* against a strong CLIP-based baseline (ViT-L/14), which computes the cosine similarity between the question and every frame, selecting the top-128 frames. As shown in Table 8, *ViTL* consistently yields higher accuracy (e.g., 63.3% vs. 58.1% on LongVideoBench), demonstrating that query-conditioned parametric selection provides essential evidence that zero-shot semantic similarity misses. **Performance Bounds: Random vs. Oracle Zooming.** To quantify the headroom for Stage 1 localization, we establish lower bounds (random sampling) and upper bounds (oracle spans). Table 9 shows that *ViTL* sig-

nificantly outperforms random zooming and closes a large portion of the gap toward the oracle upper bound (e.g., capturing over half the headroom on LongVideoBench). This confirms gains are attributable to accurate temporal localization.

Table 8. Parametric (ViTL) vs. Non-parametric (CLIP) selection.

Method	Strategy	LVideoB	LVBench
CLIP Top- <i>k</i>	Non-param.	58.1	40.3
ViTL	Parametric	63.3	47.4

Table 9. Performance bounds analysis: Random vs. Oracle.

Config	Input	LVideoB	LVBench
Lower Bound	Random	55.2	39.5
ViTL	Pred. Spans	63.3	47.4
Upper Bound	GT Spans	70.5	51.4

5.6. Mechanism Analysis: Zooming and Multi-Span Retrieval

We further disentangle the architectural decisions that drive *ViTL*’s performance. **Impact of Visual Zooming vs. Iterative Reasoning.** To verify that gains stem from accessing high-fidelity visual details rather than simply "thinking twice," we introduce a *Refine-only* baseline (Stage 2 reuses Stage 1’s low-fps frames). As summarized in Table 10, Refine-only provides only marginal gains (56.2% vs. 56.0%), whereas *ViTL* with true Zoom achieves substantial improvements (63.3%). This proves that reallocating the visual token budget to high-resolution evidence is the primary driver of performance. **Single vs. Multi-Span Retrieval.** Allowing the retrieval of multiple disjoint spans yields a +2.3% absolute accuracy gain on LVBench (47.4% vs. 45.1% with a single-span constraint). This indicates that relevant evidence in long-form videos is frequently scattered, validating our multi-span design.

Table 10. Mechanism ablation (Zoom vs. Refine). Iterative reasoning alone (Refine-only) yields negligible gains; improvements stem from high-fidelity visual zooming.

Method	Mechanism	LVideoBench	LVBench
Single Stage	Direct answer	56.0	45.3
Refine-only	Think twice (low FPS)	56.2	45.7
ViTL	Think twice (Zoom)	63.3	47.4

5.7. Qualitative Analysis

To provide further insight into the behavior and capabilities of ViTL, Figure 1 illustrates representative an example of our model’s two-stage reasoning process, showcasing its ability to generate coherent thought processes, accurately ground temporal segments, and provide correct answers. Additional qualitative examples, including comparisons with baselines and failure case analyzes, are provided in the appendix A.

6. Conclusion and Discussion

We cast long-video QA as allocating a fixed token budget to verifiable evidence and introduced *Video-in-the-Loop* (ViTL), a skim→zoom pipeline that first *localizes* evidence spans and then *answers* within them. ViTL trains an interleaved span+answer output end-to-end with a group-relative objective coupling temporal IoU and answer correctness, enabling credit to flow from answers back to localization. To supply supervision, we develop event knowledge graphs based approach to turn video into span-grounded MCQA that ties each question to *ground-truth* time span(s) with same-video distractors. Together, the data and method move tokens off background, make “where” explicit, and improve QA and grounding under matched compute.

Impact Statement

This work proposes a video-in-the-loop framework that improves long-video understanding by explicitly grounding question answering in temporally localized evidence. By reallocating computation toward question-relevant video segments under fixed budgets, the method enhances both interpretability and efficiency of video reasoning systems. Such improvements can benefit applications that require reliable and auditable video understanding, including educational assistants, embodied AI, and long-form content analysis.

The approach builds on existing multimodal large language models and standard video datasets, and does not introduce new sensing modalities, data sources, or deployment settings that would raise novel ethical concerns beyond those already associated with video understanding and generation models. As with prior work, potential risks include misuse of automated video analysis or over-reliance on model

outputs without human verification.

From a sustainability perspective, the proposed two-stage “skim→zoom” design aims to reduce unnecessary computation on irrelevant video content, offering a more compute-efficient alternative to uniformly processing long videos. By enabling better performance under constrained token and frame budgets, the method may help lower the computational and environmental costs of training and deploying long-video models.

Overall, we do not identify additional broader societal or ethical impacts specific to this work beyond those commonly discussed in the context of multimodal AI systems.

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A. More Evaluation results 7B models

We further enhance Qwen2.5-VL-7B model with our two-stage RL training pipeline *ViTL* and compare the results on temporal grounding task and QA answering task on CharadeSTA (Gao et al., 2017) and CG-Bench (Chen et al., 2024a).

Charades-STA (Gao et al., 2017). As presented in Table 11, *ViTL (Ours 7B)* achieves strong performance, outperforming several specialized methods in mIoU and recall at various IoU thresholds. The *ViTL (Ours 3B)* variant also shows competitive results.

CG-Bench (Chen et al., 2024a). Table 12 shows that *ViTL (Ours 3B)* and *ViTL (Ours 7B)* achieves competitive mIoU for grounding compared to other models of similar and larger sizes, while also providing a strong long-form accuracy.

B. Additional Analyses and Discussions

B.1. Robustness to Temporal Grounding Bias and QVHighlights Transfer

A recent line of work shows that temporal grounding models often exploit dataset-specific priors (e.g., typical moment location and duration) instead of learning true cross-modal reasoning. To verify that *ViTL* does not rely on such priors, we evaluate it on ActivityNet-CD, an out-of-distribution (OOD) split of ActivityNet-Captions specifically constructed to break these biases, and further perform zero-shot moment retrieval on QVHighlights.

As shown in Table 13, *ViTL (7B)* maintains strong performance under distribution shift: its mIoU drops only slightly from 24.1 to 23.3 when moving from the in-domain ActivityNet-Captions split to ActivityNet-CD, while still clearly outperforming the Qwen2.5-VL-7B baseline in both settings. On QVHighlights zero-shot transfer, *ViTL* also improves mAP@5 by +1.3 absolute (14.2 vs. 12.9), supporting that our “Skim–Zoom” pipeline learns query-conditioned grounding rather than memorizing temporal priors.

B.2. Open-Ended QA Capabilities

While our main experiments adopt multiple-choice QA for standardized evaluation, *ViTL* is built on top of generalist MLLMs (Qwen2.5-VL) and naturally supports open-ended generation. To assess these generative capabilities, we evaluate *ViTL* on EgoTempo, an open-ended QA benchmark.

Table 14 shows that *ViTL* improves open-ended QA accuracy from 26.1% (Qwen2.5-VL-7B) to 31.0%, demonstrating that our “Skim–Zoom” temporal grounding also benefits free-form reasoning and not only multiple-choice formats.

B.3. Latency and Resource Cost

Token-budget matching does not fully capture inference efficiency because *ViTL* performs a two-stage skim–zoom pass. We therefore measure end-to-end latency on a single A100-80GB GPU with 128 input frames and matched total visual tokens. As shown in Table 15, *ViTL* improves ac-

curacy under the same token budget but introduces modest overhead from the second prefill and span parsing.

C. More Qualitative Results

We provide additional qualitative examples of our video-in-the-loop approach in Figure 3, Figure 4, Figure 5 and Figure 6

D. Declaration of LLM Usage

We used large language models (LLMs) solely for light editing of prose—including wording refinement, grammar correction, and minor clarity improvements—in limited portions of this paper. All LLM-edited text was subsequently reviewed and revised by the authors.

Table 11. Zero-shot Video Temporal Grounding on Charades-STA (Gao et al., 2017). **Bold** indicates our model’s results and its scores.

Method	Size	R@0.3 (%) ↑	R@0.5 (%) ↑	R@0.7 (%) ↑	mIoU (%) ↑
VTimeLLM (Huang et al., 2024)	13B	55.3	34.3	14.7	34.6
TimeChat (Ren et al., 2023)	7B	51.5	32.2	13.4	–
Momentor (Qian et al., 2024)	7B	42.6	26.6	11.6	28.5
HawkEye (Zhao et al., 2024)	7B	50.6	31.4	14.5	33.7
ChatVTG (Qu et al., 2024)	7B	52.7	33.0	15.9	34.9
VideoChat-TPO (Yan et al., 2024)	7B	58.3	40.2	18.4	38.1
E.T. Chat (Liu et al., 2024b)	4B	65.7	45.9	20.0	42.3
ViTL (Ours 3B)	3B	77.7	63.5	36.3	54.0
ViTL (Ours 7B)	7B	80.1	66.0	40.2	59.0

 Table 12. Grounded VideoQA performance on CG-Bench. Closed-source APIs (top) and open-source models (bottom) are grouped separately and sorted by parameter size. **Bold** indicates our model’s results and its scores. ViTL (3B/7B) consistently outperforms open-source baselines of similar and larger sizes in grounding quality (mIoU).

Method	Size	Type	long-acc. (%) ↑	mIoU (%) ↑
<i>Closed-source (API)</i>				
Gemini-1.5-Flash (Team et al., 2024)	–	API	32.3	3.67
GPT-4o-mini (OpenAI, 2024)	–	API	33.4	3.75
Gemini-1.5-Pro (Team et al., 2024)	–	API	37.2	3.95
Claude-3.5-Sonnet (Anthropic, 2024)	–	API	40.5	3.99
GPT-4o (OpenAI, 2024)	–	API	45.2	5.62
<i>Open-source models</i>				
Qwen2.5VL-instruct (Team, 2025)	3B	Open	18.4	0.86
ViTL (Ours 3B)	3B	Open	23.5	2.90
Video-LLaVA (Lin et al., 2023)	7B	Open	16.2	1.13
VideoLLaMA 2 (Zhang et al., 2023)	7B	Open	18.4	1.21
Videochat2 (Li et al., 2023)	7B	Open	19.3	1.28
Qwen-VL-Chat (Bai et al., 2023)	7B	Open	21.6	0.89
ST-LLM (Liu et al., 2023)	7B	Open	23.8	2.23
LongVA (Zhang et al., 2024a)	7B	Open	28.7	2.94
LLaVA-OV (Li et al., 2024a)	7B	Open	31.1	1.63
ViTL (Ours 7B)	7B	Open	34.4	3.32
MiniCPM-v2.6 (Yao et al., 2024)	8B	Open	30.1	2.35
Kangaroo (Liu et al., 2024a)	8B	Open	30.2	2.56
Chat-UniVi-v1.5 (Jin et al., 2023)	13B	Open	25.9	2.07
Video-CCAM (Fei et al., 2024)	14B	Open	29.7	2.63
ShareGPT4Video (Chen et al., 2024b)	16B	Open	26.7	1.85
VITA (Fu et al., 2024)	8×7B	Open	33.3	3.06
Qwen2-VL (Wang et al., 2024)	72B	Open	41.3	3.58
InternVL2 (OpenGVLab Team, 2024)	78B	Open	42.2	3.91

Table 13. In-domain vs. distribution-shift generalization and zero-shot transfer. ViTL maintains strong performance on ActivityNet-CD (OOD) and improves zero-shot moment retrieval on QVHighlights.

Dataset	Setting	Metric	Qwen2.5-VL-7B	ViTL (7B)
ActivityNet-Captions	In-domain (Train = Test dist.)	mIoU	22.5	24.1
ActivityNet-CD	OOD (distribution-shift)	mIoU	20.2	23.3
QVHighlights	Zero-shot moment retrieval	mAP@5	12.9	14.2

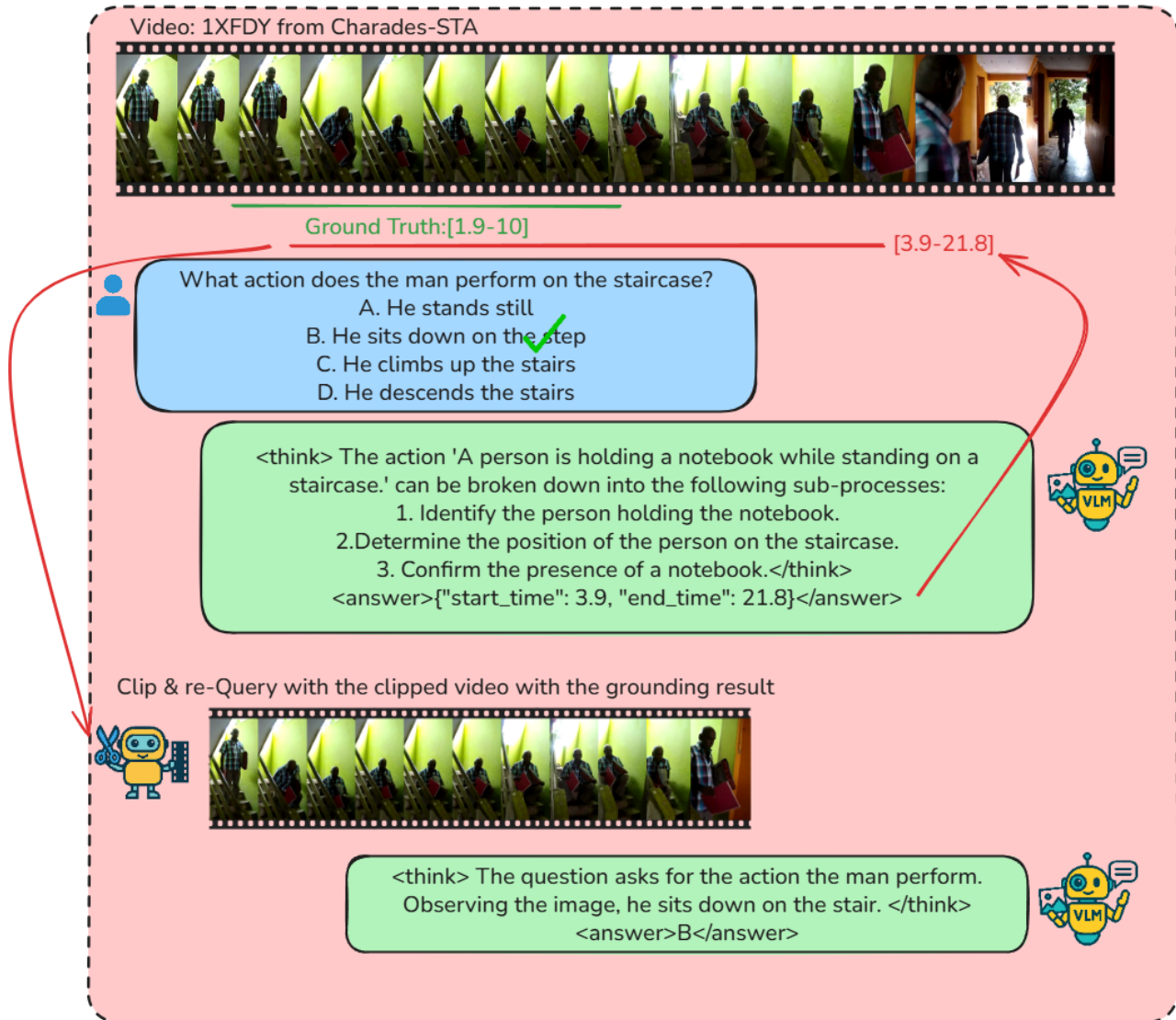


Figure 3. Qualitative demonstration of our two-stage reasoning and grounding pipeline on a sample video from Charade-STA.

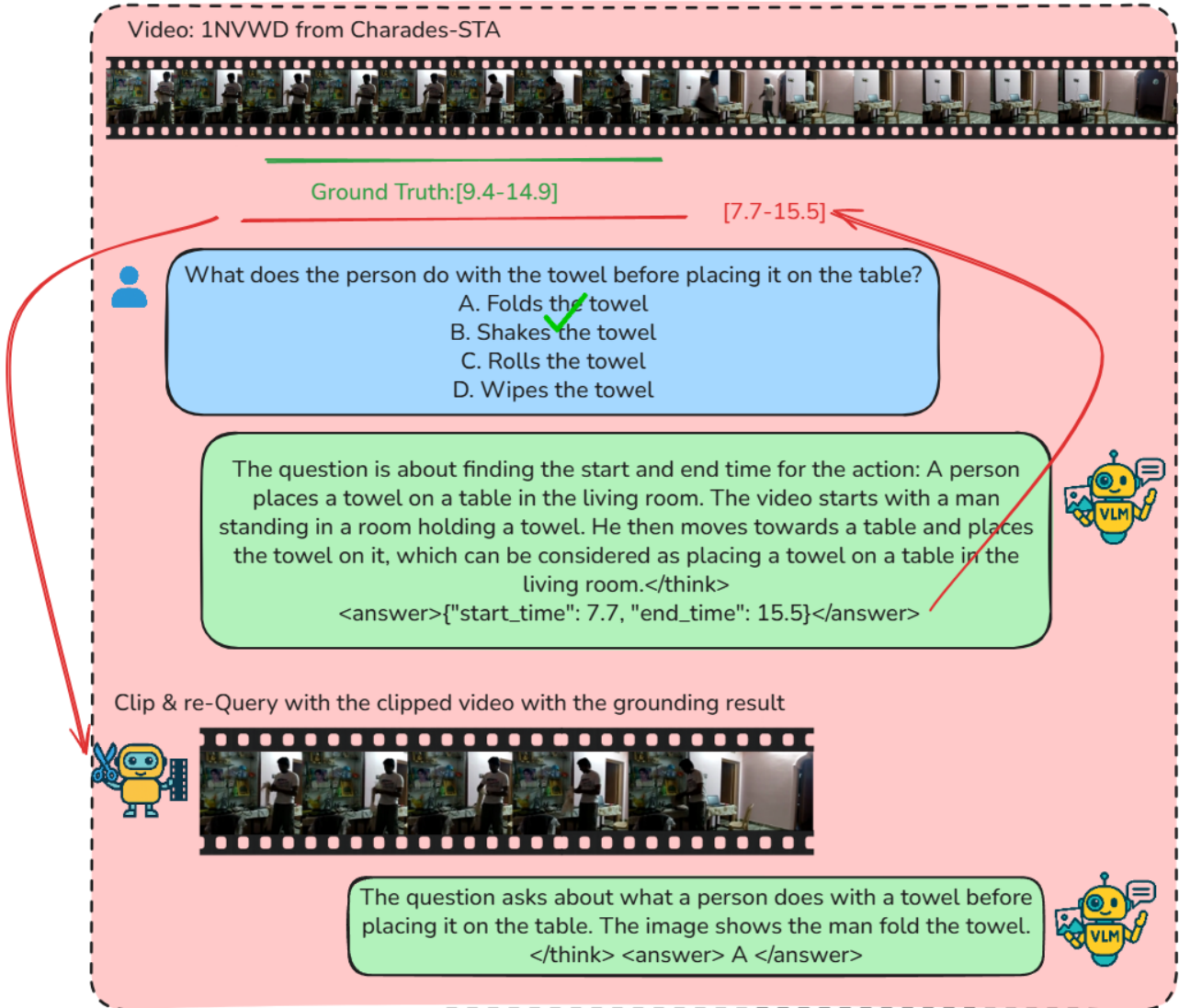


Figure 4. Qualitative demonstration of our two-stage reasoning and grounding pipeline on a sample video from Charade-STA.

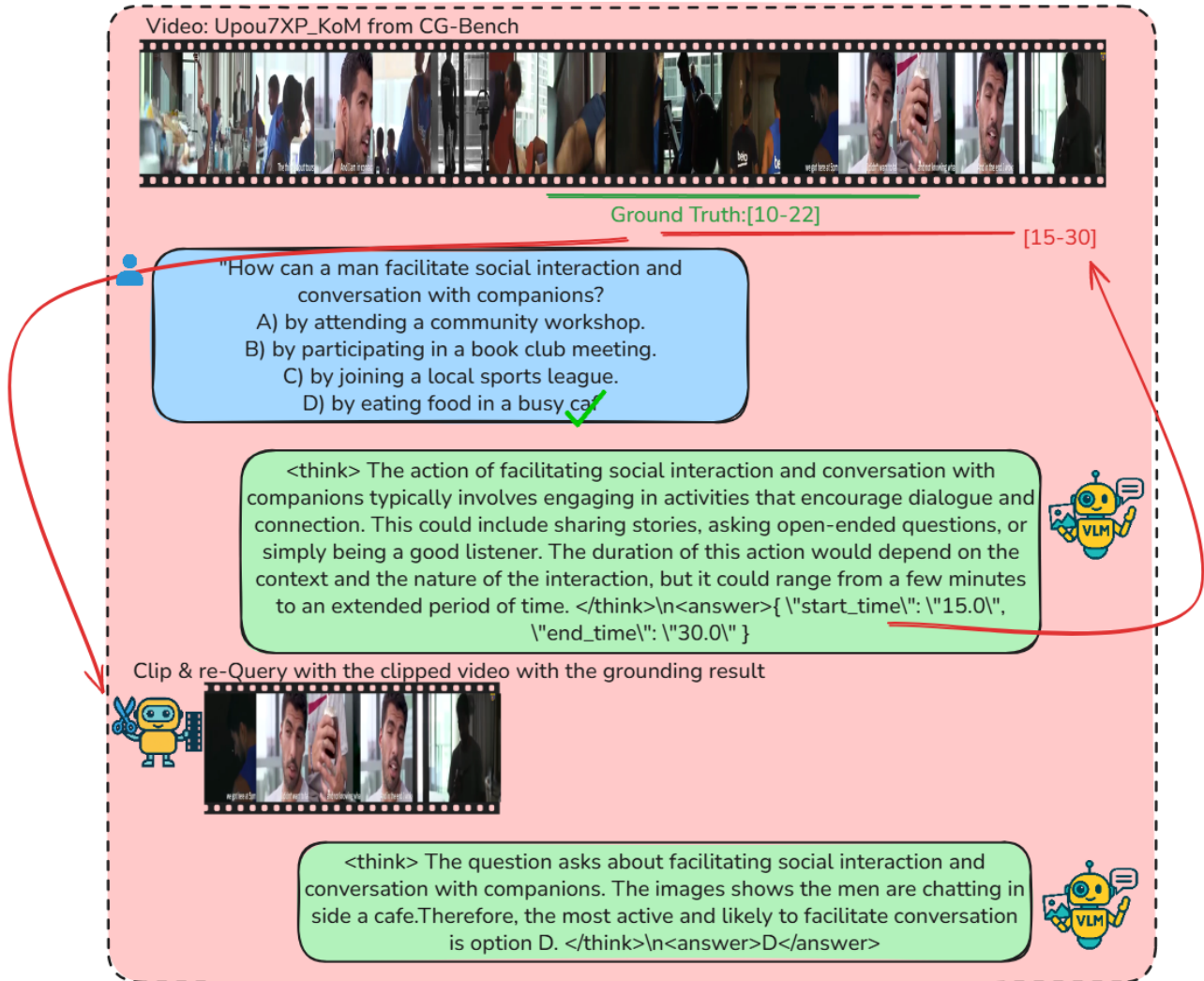


Figure 5. Qualitative demonstration of our two-stage reasoning and grounding pipeline on a sample video from CG-Bench.

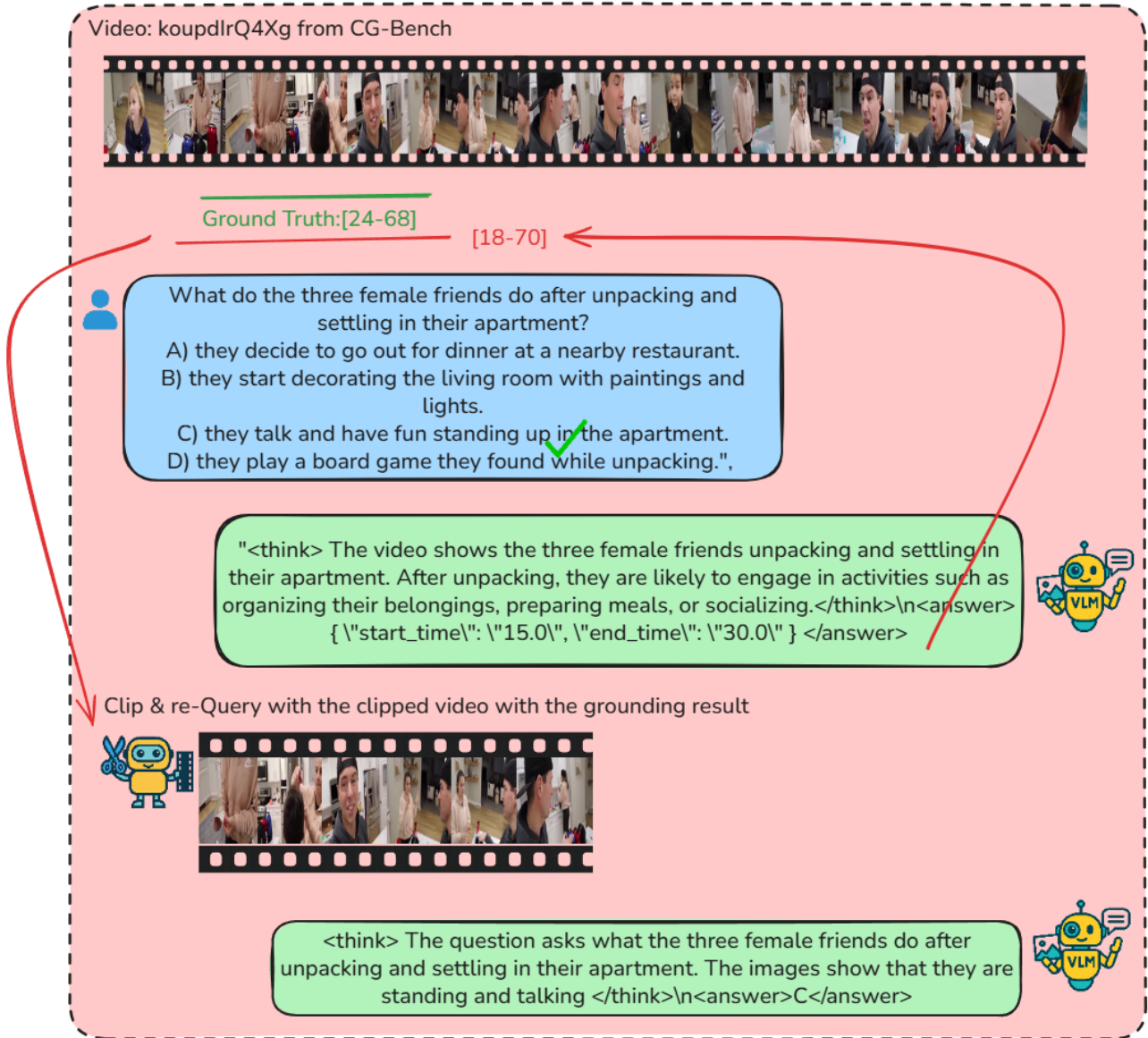


Figure 6. Qualitative demonstration of our two-stage reasoning and grounding pipeline on a sample video from CG-Bench.

Table 14. Open-ended QA performance on EgoTempo. *ViTL* improves open-ended QA accuracy over the Qwen2.5-VL-7B backbone.

Model	Task	Accuracy (%)
Qwen2.5-VL-7B	Open-ended QA	26.1
<i>ViTL</i> (Ours)	Open-ended QA	31.0

Table 15. Latency and memory under matched token budget.

Method	Total tokens	Latency (s)	Peak mem. (GB)
Baseline (448px)	16,413	6.24	21.74
<i>ViTL</i> (Ours)	16,413	7.32	22.05